Donovan Crowley

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Education

Case Western Reserve University **Cleveland, OH**
- B.S.E. in Electrical Engineering & B.S. in Computer Science - GPA: 4.0 Aug. 2024 - May 2028
- Relevant Coursework: Data Structures, Circuits, Logic Design, Discrete Math, Algorithms, Operating Systems
- Qualified as a top 50 programmer in the 2025 CWRU RFB Programming Competition April 2025

Avon Old Farms, Avon, CT - GPA: 4.2/4.3 (Valedictorian) Sept. 2020 - May 2024
- Colatella Family Scholarship
- Harvard Book Award
- AP Scholar with Distinction

Professional Experience

SDLE Undergraduate Researcher (Paid) 2025 - Present
- Working with graduate students and faculty on building neural networks and machine learning models for computer vision analysis

Titan Pool Services Inc., Head Lifeguard with Pool Operating License 2021 - 2024
- Built a Remotely Operated Vehicle (ROV) to clean pools, improving efficiency and encouraging service
- Supervised a team of 6 lifeguards, ensuring safety and enhancing customer satisfaction for 150+ daily patrons
- Repaired integral pool equipment such as pipes, VS pumps, and chlorinators by identifying faults, performing priming procedures, and restoring functionality to minimize customer disturbances

Projects

VTOL Drone
- Developed computer vision with a Hough Circle Transform detection algorithm using an ESP32 camera for object detection and precise targeting
- Built six circuit boards to connect the Pixhawk to each of the motors on the drone
- Team placed first at the 2025 VFS Design-Build-Vertical-Flight Competition
- *Utilized*: C++, Python, ESP32, Raspberry Pi

Web Scraper
- Built a web scraper to extract basic information from multiple websites' APIs
- Used transformers built on Hugging Face to determine cosine similarities of Wikipedia page links in an attempt to find the shortest path between the unrelated websites
- *Utilized*: Node.js, Python, Puppeteer, Cheerio, Axios

Object Detection
- Implemented a real-time object tracking algorithm that searches for circles using the Hough Circle Transform detection algorithm and analyzes HSV values to track a ball
- Features a tracking mode that applies a red line behind the object's path that clears and resets when instructed to
- *Utilized*: OpenCV, C++

Electric Vehicle + Data Monitor
- Improved the vehicle's performance by redesigning and optimizing the circuit to improve reliability and functionality
- Incorporated a speedometer and multimeter to display the speed of the car and the voltage remaining in the batteries, improving race performance
- Team placed third at the Lime Rock Park Electrathon Competition
- *Utilized:* Multimeter, Autodesk Fusion 360

Remotely Operated Vehicle
- Conceptualized, 3D modeled, and fabricated a PVC robot equipped with a waterproof circuit and filter system to collect debris from the pool floor
- *Utilized*: Python, Arduino, SolidWorks

Skills

Programming Languages: Python, C++, Java, Node.js, HTML, CSS, JavaScript, MATLAB, Arduino
Software: Autodesk Fusion 360, Origin, SolidWorks